Investigating the Interaction Between SAR Normalization and Data Augmentation for Cross-Regional Generalization in Sentinel-1 Water Segmentation

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Abstract

Deep learning models for Synthetic Aperture Radar (SAR) surface-water segmentation often lose accuracy when deployed on geographically unseen flood events. Input normalization and data augmentation are routine preprocessing steps, yet how they interact under geographical distribution shift is poorly understood. This study examines how four SAR normalization strategies (unscaled raw decibels, Min-Max scaling, Z-score standardization, and percentile clipping with standardization) interact with geometric and radiometric intensity augmentations across ten controlled U-Net experiments on 446 hand-labeled Sentinel-1 chips from the Sen1Floods11 dataset. Models are trained on five source regions, validated on three, and tested on both held-out in-distribution (ID) samples and three unseen out-of-distribution (OOD) regions: Bolivia, Mekong and Spain. The results show a crossover interaction between normalization and geometric augmentation. Under Min-Max scaling, geometric augmentation raises OOD Intersection over Union (IoU) by 5.11 percentage points (0.6889 to 0.7400); under Z-score standardization, the same augmentation lowers overall OOD IoU by 2.41 percentage points (0.7771 to 0.7530) because of false positives in Spain. Keeping native decibel values without augmentation yields the highest overall OOD IoU (0.7837), while combining percentile clipping with intensity augmentation achieves the best IoU in Bolivia (0.7313). Backscatter analysis shows that regional accuracy depends on water-to-land contrast and on the overlap of dark land with the water backscatter range.

Keywords: Synthetic Aperture Radar, Flood Segmentation, Domain Generalization, Data Normalization, U-Net

1. Introduction

Accurate delineation of surface-water extent during flood events is essential for emergency response, damage assessment and hydrological modeling [1], [2]. Optical satellite sensors such as Sentinel-2 and Landsat-8 provide high-resolution multispectral imagery, but persistent cloud cover, heavy precipitation and solar illumination requirements restrict their availability during active flooding [1], [9]. Synthetic Aperture Radar (SAR) sensors, particularly the European Space Agency's C-band (5.405 GHz) Sentinel-1 mission, avoid these constraints by transmitting microwave pulses that penetrate clouds and operate day and night [1], [2]. Calm open water acts as a specular reflector that directs radar energy away from the side-looking sensor, producing low backscatter returns, whereas rough terrain, vegetation and urban structures scatter energy back toward the sensor and give higher backscatter values [2], [9].

Despite this physical contrast, automated SAR flood segmentation is complicated by coherent speckle noise, variations in incidence angle, wind-roughened water surfaces and topographic shadow or layover effects [2], [8]. While Convolutional Neural Networks (CNNs) such as U-Net [3] outperform classical Otsu histogram thresholding [10] on benchmark datasets like Sen1Floods11 [1], models evaluated on random image splits suffer substantial accuracy drops when tested on geographically unseen flood events [4], [5], [6]. Differences in soil moisture, vegetation canopy structure and terrain slope across hydro-climatic regimes cause models to overfit the backscatter distributions of their source domains [4], [6].

In existing SAR flood-mapping studies, preprocessing choices such as decibel (dB) clipping, Min-Max scaling, Z-score standardization, percentile clipping and Lee speckle filtering [8] are typically combined heuristically with spatial and radiometric augmentations [2], [7], [11]. It remains unclear whether normalization and augmentation act additively or interact non-linearly across distinct geographical environments. Rotating or flipping side-looking SAR imagery, for example, changes the orientation of radar shadows in pixel coordinates relative to the sensor look direction, while synthetic intensity perturbations can push unclipped radar outliers into unrealistic numerical ranges.

This study evaluates the interaction between SAR normalization strategies and data augmentation regimes on the 446 hand-labeled Sentinel-1 chips of the Sen1Floods11 benchmark [1]. To isolate preprocessing effects, the U-Net architecture, training budget, masked loss function and geographical data splits are held constant across ten selected experimental configurations (E0 through E8, plus a factorial Min-Max geometric variant). Because an exhaustive 16-cell factorial grid was too costly to run, we evaluate a targeted ten-experiment matrix that isolates the main normalization effects, the main augmentation effects and selected crossover interactions. This work makes four contributions:

1. A controlled comparison of four SAR input representations (raw dB, Min-Max scaling, Z-score standardization and 1st–99th percentile clipping with standardization), evaluated on both familiar in-distribution (ID) and unseen out-of-distribution (OOD) flood events.
2. An isolated evaluation of geometric, radiometric intensity and combined augmentations, alongside a supplementary 5×5 Lee speckle filtering baseline [8], on cross-regional generalization.
3. Identification of a crossover interaction between normalization and augmentation: geometric augmentation improves uncentered Min-Max models (+5.11 percentage points OOD IoU) but degrades zero-centered Z-score models in Spain (-6.82 percentage points IoU), whereas percentile clipping stabilizes radiometric intensity augmentation in Bolivia (+9.23 percentage points IoU relative to unclipped Z-score intensity augmentation).

4. A quantitative backscatter analysis across all 11 Sen1Floods11 events linking regional segmentation errors to VV/VH water-to-land contrast (Δ dB) and dark-land histogram overlap below -15 dB.

2. Related Work

2.1 Deep Learning for SAR Flood Segmentation

Early automated SAR flood mapping relied on radiometric thresholding algorithms, such as global Otsu thresholding [10] and split-based bimodal histogram tiling [9], to separate low-backscatter water from higher-backscatter land. Single-threshold methods, however, struggle when wind roughens water surfaces or when emergent vegetation increases backscatter through double-bounce reflections [2]. Following the introduction of Fully Convolutional Networks (FCNs) [12], encoder-decoder architectures such as U-Net [3], ResNet-backed segmentation networks [13], DeepLabv3+ [14] and UNet++ [15] became the standard for pixel-level flood delineation [1], [11], [16]. Bonafilia et al. [1] introduced the globally distributed Sen1Floods11 benchmark, demonstrating that FCNs and U-Net models outperform a standard -15 dB VV backscatter threshold on both permanent-water and flood-water segmentation. Subsequent studies incorporated multi-temporal Sentinel-1 pairs [16], [17] and interferometric SAR coherence [18] to improve urban and vegetated flood detection, although single-date dual-polarization (VV/VH) segmentation is still needed for rapid emergency response when pre-flood baseline acquisitions are unavailable [1], [7].

2.2 SAR Preprocessing and Speckle Filtering

Because raw SAR intensity follows a multiplicative exponential distribution with heavy right-skewness, standard pipelines convert linear power values into logarithmic decibels (dB) [1], [2]. To suppress the granular speckle noise caused by coherent wave interference, classical adaptive spatial filters such as the Lee filter [8] compute local window statistics to smooth homogeneous areas while preserving high-variance edges. In deep learning workflows, however, there is no consensus on input normalization. Some studies apply fixed physical Min-Max clipping (e.g., $[-30, 0]$ dB scaled to $[0, 1]$) [11], others apply global Z-score standardization [1], [4], and recent frameworks use percentile Winsorization (e.g., 1st–99th percentile clipping) prior to standardization to suppress extreme double-bounce urban spikes [7]. No study has systematically isolated whether convolutional BatchNormalization layers [19] make external scaling redundant, or whether non-zero-centered scaling interacts adversely with spatial augmentations.

2.3 Domain Generalization Across Flood Events

Recent literature shows that random chip-level train-test splits overestimate model accuracy because of spatial autocorrelation among tiles from the same flood event [4], [5], [6]. Turkmenli et al. [4] evaluated domain generalization methods across individual SAR flood events treated as distinct source domains, reporting IoU gains of roughly 3 percentage points on unseen events. Dhagat [5] compared random splits against country-level geographical holdouts on Sentinel-1 and Sentinel-2 imagery, showing that a vanilla U-Net IoU drops from 0.727 under random splitting to 0.464 under country-level holdouts. Similarly, Ratté-Fortin et al. [6] investigated leave-one-location-out spatial transferability across 18 flood events, with attention to the role of hydro-climatic and geomorphological similarity between training and target regions. Adhikari et al. [7] combined percentile-clipped VV/VH inputs with Joint Research Centre (JRC) permanent-water priors in a UNet++ framework evaluated on held-out events in Bolivia and Nepal. While these studies confirm the severity of geographic domain shift, they hold preprocessing and augmentation fixed; the present work isolates how normalization and augmentation choices interact across held-out regions.

3. Methodology

3.1 Dataset and Geographical Splitting

Experiments use the hand-labeled Sentinel-1 subset (S1Hand and LabelHand) of the Sen1Floods11 (v1.1) dataset [1]. Although Sen1Floods11 includes 4,831 weakly labeled chips derived from optical/radar thresholding, those weak labels contain classification noise that can distort cross-regional evaluation [1]. We therefore restrict all training, validation and testing to the 446 expert hand-labeled chips. Each chip is a 512×512 pixel dual-polarization SAR image (two channels: VV, vertical transmit/vertical receive, and VH, vertical transmit/horizontal receive) at 10 m spatial resolution, paired one-to-one with a ground-truth mask. Pixel intensities are provided in calibrated logarithmic decibels (dB). Ground-truth masks encode three integer values: 1 for surface water (floodwater and permanent water), 0 for non-water land, and -1 for invalid pixels along satellite swath borders.

To evaluate both familiar-region performance and unseen cross-regional transferability [5], the 11 flood events are partitioned into four mutually exclusive subsets:

1. **Training set (familiar regions, 256 chips):** sampled from five source events: USA (59), India (58), Paraguay (57), Ghana (46) and Sri Lanka (36).
2. **In-distribution (ID) test set (familiar regions, 43 chips):** held out deterministically by sorting chip filenames alphanumerically within the five familiar regions (familiar_all[::7]) and selecting every seventh chip (USA: 10, India: 10, Paraguay: 10, Ghana: 7, Sri Lanka: 6). Because chips from the same flood event come from the same regional satellite overpass, ID test chips may exhibit regional spatial correlation with training chips; ID IoU is therefore used strictly as a familiar-domain reference, while primary generalization claims rely on the geographically disjoint OOD test set.
3. **Validation set (model selection, 72 chips):** composed of three independent events, Somalia (26), Pakistan (28) and Nigeria (18), used exclusively for checkpoint selection and early stopping.
4. **Out-of-distribution (OOD) test set (unseen regions, 75 chips):** composed of three geographically distinct held-out events across three continents: Bolivia (15 chips; tropical Amazonian floodplain), Mekong (30 chips; Southeast Asian monsoonal delta) and Spain (30 chips; Mediterranean agricultural and semi-arid terrain).

3.2 SAR Preprocessing and Invalid-Pixel Handling

SAR segmentation is sensitive to the order of operations between invalid-pixel masking and numerical normalization. In raw_db, 0.0 dB represents an intense radar return; in minmax_clip, 0.0 represents the -30 dB floor; and in zscore, 0.0 represents the mean backscatter. To prevent invalid swath borders from distorting spatial filters or convolutional activations, each chip is processed in four sequential steps:

1. **Validity mask construction:** a binary validity mask $V \in \{0,1\}^{512\times512\times1}$ is defined prior to any transformation, as given in Eq. (1), where $B \in \{\mathrm{VV},\mathrm{VH}\}$.
2. **Leak-free training statistics:** global normalization parameters are computed strictly over valid pixels ($V_{i,j}=1$) in the familiar training regions: $\mu_{\mathrm{VV}} = -10.1397$ dB, $\sigma_{\mathrm{VV}} = 3.8141$ dB, 1st percentile $P_{01,\mathrm{VV}} = -22.50$ dB and 99th percentile $P_{99,\mathrm{VV}} = -1.50$ dB; for VH, $\mu_{\mathrm{VH}} = -16.9252$ dB, $\sigma_{\mathrm{VH}} = 4.3560$ dB, $P_{01,\mathrm{VH}} = -29.50$ dB and $P_{99,\mathrm{VH}} = -7.50$ dB.
3. **Pre-filter imputation:** invalid pixels ($V_{i,j}=0$) are temporarily imputed with μ_B so that NaN values do not corrupt the 5×5 sliding-window variance calculations during Lee filtering.
4. **Post-normalization zero-clamping:** immediately after applying the chosen normalization formula (and after any intensity augmentation), all invalid pixels are clamped to $X_{i,j}=0.0$. Although $V_{i,j}$ is not passed as a third input channel to the U-Net encoder, clamping $X_{i,j}=0.0$ ensures a zero multiplicative weight contribution in the first bias-free 3×3 convolution (use_bias=False), and $V_{i,j}$ is passed directly to the loss and evaluation functions to exclude invalid pixels from all gradient updates and metrics.

$$V_{i,j} = \begin{cases} 1, & \text{if } y_{i,j} \neq -1 \text{ and } B_{i,j} \notin \{\mathrm{NaN}, \pm\infty\} \text{ for } B \in \{\mathrm{VV},\mathrm{VH}\} \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

3.2.1 Normalization Formulas

Four input representations are evaluated for each band $B \in \{\mathrm{VV},\mathrm{VH}\}$:

1. **Raw dB (raw_db):** valid intensities remain in native decibel units, as in Eq. (2).

$$B_{\mathrm{raw}} = B \tag{2}$$

2. **Min-Max scaling (minmax_clip):** values are clipped to $[-30.0, 0.0]$ dB and scaled linearly to $[0,1]$, as in Eq. (3).

$$B_{\mathrm{mm}} = \frac{\min(\max(B,-30.0),0.0)+30.0}{30.0} \tag{3}$$

3. **Z-score standardization (zscore):** values are standardized to zero mean and unit variance, as in Eq. (4).

$$B_{\mathrm{z}} = \frac{B-\mu_B}{\sigma_B} \tag{4}$$

4. **Percentile clipping with standardization (percentile):** values are Winsorized to $[P_{01,B}, P_{99,B}]$ before Z-score standardization, as in Eq. (5).

$$B_{\mathrm{pct}} = \frac{\min(\max(B,P_{01,B}),P_{99,B})-\mu_B}{\sigma_B} \tag{5}$$

3.2.2 Lee Speckle Filtering

In experiment E8, a 5×5 local Lee adaptive speckle filter [8] is applied independently to the raw VV and VH bands (in dB) prior to Z-score standardization. Using scipy.ndimage.uniform_filter with a 5×5 window, local mean $\bar{I}$, local variance $\sigma^2_{\mathrm{local}} = \max\left(\overline{I^2}-\bar{I}^2, 0\right)$ and chip-wide variance $\sigma^2_{\mathrm{global}}$, the filtered intensity $\hat{I}$ is computed as in Eq. (6).

$$\hat{I} = \bar{I} + W\left(I-\bar{I}\right), \qquad W = \frac{\sigma^2_{\mathrm{local}}}{\sigma^2_{\mathrm{local}}+\sigma^2_{\mathrm{global}}+\varepsilon} \tag{6}$$

where $\varepsilon = 10^{-8}$.

3.3 Data Augmentation Regimes and Experimental Matrix

Four training augmentation regimes are implemented:

1. **No augmentation (none):** chips are fed in their native orientation and radiometric scale.
2. **Geometric augmentation (geometric):** random horizontal flips ($p=0.5$), vertical flips ($p=0.5$) and discrete 90° rotations ($k \in \{0,1,2,3\}$ with equal probability 0.25) are applied identically to X, y and V.
3. **Intensity augmentation (intensity):** with probability $p=0.5$, a uniform calibration offset $\delta \sim \mathscr{U}(-0.15, 0.15)$ and additive zero-mean Gaussian noise $\eta \sim \mathscr{N}(0, 0.08^2)$ are added to all valid pixels ($V_{i,j}=1$) in the normalized tensor X, followed by re-clamping of invalid pixels ($V_{i,j}=0$) to 0.0.
4. **Combined augmentation (combined):** both geometric and intensity transformations are applied sequentially.

Training all 16 cells of the full factorial grid on a CPU workstation would require over 45 hours of compute, so we use a structured ten-experiment fractional design (Table 1). First, all four normalizations are evaluated under none (E0, E1, E2, E3) to isolate the main effect of normalization. Second, all four augmentation regimes are evaluated under zscore (E2, E4, E5, E6) to isolate the main effect of augmentation. Third, two targeted interaction cells, MinMax_Geometric (minmax_clip $\times$ geometric) and E7_Percentile_Intensity (percentile $\times$ intensity), test whether spatial regularization rescues uncentered Min-Max scaling and whether outlier clipping stabilizes radiometric noise injection; these are supplemented by the Lee filter baseline (E8). The remaining six cells (raw_db with augmentations, minmax_clip with intensity/combined, and percentile with geometric/combined) were excluded to prioritize these primary physical hypotheses.

3.4 U-Net Architecture and Masked Loss Formulation

All experiments use an identical 2D U-Net [3] taking a $(512,512,2)$ input tensor. The encoder comprises four stages with

Table 1. Fractional experimental matrix of evaluated (✓) and excluded (×) configurations across normalization and augmentation regimes.

Configuration ID	Normalization	Augmentation	Status
E0_RawDB_NoAug	raw_db	none	✓ evaluated
E1_MinMax_NoAug	minmax_clip	none	✓ evaluated
E2_ZScore_NoAug	zscore	none	✓ evaluated
E3_Percentile_NoAug	percentile	none	✓ evaluated
E4_ZScore_Geometric	zscore	geometric	✓ evaluated
E5_ZScore_Intensity	zscore	intensity	✓ evaluated
E6_ZScore_Combined	zscore	combined	✓ evaluated
E7_Percentile_Intensity	percentile	intensity	✓ evaluated
E8_ZScore_LeeFilter	zscore	none (5×5 Lee)	✓ evaluated
MinMax_Geometric	minmax_clip	geometric	✓ evaluated
raw_db × geometric / intensity / combined	raw_db	geometric, intensity, combined	× excluded
minmax_clip × intensity / combined	minmax_clip	intensity, combined	× excluded
percentile × geometric / combined	percentile	geometric, combined	× excluded

Ten of the sixteen cells of the full normalization × augmentation factorial were trained; the six remaining cells listed above were excluded to prioritize the primary physical hypotheses.

32, 64, 128 and 256 filters connected by 2×2 max-pooling layers, followed by a 512-filter bottleneck. Each convolutional block consists of two 3×3 bias-free convolutions (padding="same", use_bias=False), each followed by BatchNormalization [19] and a ReLU activation. The symmetric decoder upsamples feature maps via 2×2 nearest-neighbor interpolation, concatenates skip connections and applies twin 3×3 convolutional blocks. A 1×1 sigmoid convolution outputs water probabilities $\hat{y} \in (0,1)^{512\times512\times1}$ (7,759,521 total parameters).

To handle class imbalance [20] and exclude invalid swath borders ($V_{i,j} = 0$), we optimize a custom masked binary cross-entropy (BCE) plus soft Dice loss, defined in Eqs. (7)-(9).

$$\mathscr{L}_{\text{BCE}} = -\frac{\sum_{i,j} V_{i,j} \left[y_{i,j} \ln(\hat{y}_{i,j}) + (1 - y_{i,j}) \ln(1 - \hat{y}_{i,j}) \right]}{\sum_{i,j} V_{i,j} + \varepsilon} \tag{7}$$

$$\mathscr{L}_{\text{Dice}} = 1 - \frac{2 \sum_{i,j} V_{i,j}\, y_{i,j}\, \hat{y}_{i,j} + \varepsilon_d}{\sum_{i,j} V_{i,j} \left(y_{i,j} + \hat{y}_{i,j} \right) + \varepsilon_d} \tag{8}$$

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{BCE}} + \mathscr{L}_{\text{Dice}} \tag{9}$$

where $\hat{y}_{i,j}$ is clipped to $[10^{-7}, 1 - 10^{-7}]$, $\varepsilon = 10^{-7}$ and $\varepsilon_d = 10^{-5}$.

3.5 Hyperparameters, Convergence and Evaluation Metrics

Models are implemented in Python 3.10 using TensorFlow/Keras 2.13, NumPy 1.24, SciPy 1.11 and Rasterio 1.3 on an Intel Core i5-7500T CPU workstation (2.70 GHz). Training uses an in-memory sequence generator (batch_size=2, 128 steps/epoch), the Adam optimizer [21] (lr $= 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$), a fixed random seed (seed=42), a 10-epoch budget (≈ 3.2 hours/model), ModelCheckpoint saving the weights that maximize batch-averaged validation masked IoU (val_masked_iou, mode="max"), and EarlyStopping(patience=5, restore_best_weights=True). As Table 2 shows, validation IoU peaked between epoch 2 and epoch 10 across all configurations (epoch 2 for E4; epoch 6 for E0, MinMax_Geometric, E3, E6 and E7; epoch 8 for E5 and E8; epoch 10 for E1 and E2), which shows that 10 epochs (1,280 gradient updates) are sufficient for validation convergence on the 256-chip training set.

During evaluation, probability maps are thresholded at $\hat{y}_{i,j} > 0.5$. Global pixel-level true positives (TP), false positives (FP) and false negatives (FN) are accumulated strictly over valid pixels ($V_{i,j} = 1$) to compute dataset-level Intersection over Union (IoU), Dice coefficient, precision, recall and the ID-to-OOD difference ($\Delta\text{IoU}_{\text{gap}}$), as defined in Eqs. (10)-(14).

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{10}$$

$$\text{Dice} = \frac{2TP}{2TP + FP + FN} \tag{11}$$

$$\text{Prec} = \frac{TP}{TP + FP} \tag{12}$$

$$\text{Rec} = \frac{TP}{TP + FN} \tag{13}$$

$$\Delta\text{IoU}_{\text{gap}} = \text{IoU}_{\text{ID}} - \text{IoU}_{\text{OOD}} \tag{14}$$

To quantify dark-land shadow confusion in our backscatter analysis, we also define the exploratory dark-land overlap percentage ($\text{DLO}_{<-15\,\text{dB}}$) as the percentage of valid non-water land pixels ($y_{i,j} = 0$, $V_{i,j} = 1$) whose VV backscatter falls below the classical -15.0 dB water threshold [1], as given in Eq. (15),

$$\text{DLO}_{<-15\,\text{dB}} = 100 \times \frac{N_{\text{dark}}}{N_{\text{land}}} \tag{15}$$

Here N_{land} is the number of valid non-water land pixels ($V_{i,j} = 1$, $y_{i,j} = 0$) and N_{dark} is the number of those pixels whose VV backscatter is below -15.0 dB.

Table 2. Detailed hyperparameter configuration and best validation checkpoint epochs across all ten experiments.

Exp. ID	Norm.	Lee filter	Aug. mode	Train chips	Best epoch	Peak val_masked_iou	Min val_loss
E0_RawDB_NoAug	raw_db	False	none	256	6	0.2650	0.9669
E1_MinMax_NoAug	minmax_clip	False	none	256	10	0.2458	1.0740
E2_ZScore_NoAug	zscore	False	none	256	10	0.2766	0.9512
E3_Percentile_NoAug	percentile	False	none	256	6	0.2695	0.9576
E4_ZScore_Geometric[a]	zscore	False	geometric	299[a]	2	0.2747	0.9685
MinMax_Geometric[a]	minmax_clip	False	geometric	299[a]	6	0.2531	1.0036
E5_ZScore_Intensity	zscore	False	intensity	256	8	0.2724	0.9401
E6_ZScore_Combined	zscore	False	combined	256	6	0.2756	1.2545
E7_Percentile_Intensity	percentile	False	intensity	256	6	0.2717	0.9796
E8_ZScore_LeeFilter	zscore	True (5×5)	none	256	8	0.2666	0.9476

[a]Pilot models E4_ZScore_Geometric and MinMax_Geometric were trained on all 299 familiar-region chips before the 43-chip ID holdout subset was isolated; their OOD metrics on the 75 unseen test chips (Bolivia, Mekong, Spain) are strictly held-out and leak-free, whereas their ID IoU reflects familiar-sample training evaluation.

4. Experimental Results

4.1 Quantitative Segmentation Matrix

Table 3 presents the complete quantitative evaluation across all ten experiments, ordered by unseen OOD IoU across Bolivia, Mekong and Spain.

4.2 Normalization, Augmentation and Interaction Results

Among the four unaugmented normalization baselines (E0, E1, E2, E3), E0_RawDB_NoAug achieves the highest overall unseen generalization (OOD_IoU = 0.7837, OOD_Dice = 0.8788), the highest OOD precision (0.8965) and the highest accuracy in Spain (0.7174 IoU). E2_ZScore_NoAug achieves the highest held-out in-distribution accuracy (ID_IoU = 0.5131, evaluated on the eight models trained on the 256-chip split) and the highest Mekong IoU (0.8656). E3_Percentile_NoAug (OOD_IoU = 0.7760) improves over E2 in Bolivia by +1.99 percentage points (0.6955 vs. 0.6756) and in Spain by +0.37 percentage points (0.6988 vs. 0.6951). Conversely, the unaugmented E1_MinMax_NoAug performs worst across both familiar ID chips (0.4031 IoU) and unseen OOD events (0.6889 IoU, a decrease of 9.48 percentage points relative to E0).

Comparing augmentation regimes on Z-score inputs (E2, E4, E5, E6) and Lee filtering (E8) shows opposing regional effects (Fig. 1). Adding geometric flips and rotations (E4_ZScore_Geometric) or combined augmentations (E6_ZScore_Combined) raises OOD recall from 0.8847 (E2) to 0.9067 (E4) and 0.9276 (E6), improving Bolivia IoU by +5.17 percentage points (0.6756 $\rightarrow$ 0.7273) and +3.81 percentage points (0.6756 $\rightarrow$ 0.7137), respectively. However, OOD precision drops from 0.8647 (E2) to 0.8162 (E4) and 0.7754 (E6), reducing Spain IoU by -6.82 percentage points (0.6951 $\rightarrow$ 0.6269) and -10.19 percentage points (0.6951 $\rightarrow$ 0.5932). Likewise, 5×5 Lee filtering (E8_ZScore_LeeFilter) increases Bolivia IoU by +4.70 percentage points (0.6756 $\rightarrow$ 0.7226) while reducing Mekong IoU by -2.72 percentage points (0.8656 $\rightarrow$ 0.8384) and Spain IoU by -6.08 percentage points (0.6951 $\rightarrow$ 0.6343).

As shown in Fig. 1(a), normalization and augmentation show two distinct interactions:

1. **Normalization $\times$ geometric interaction:** under minmax_clip, adding geometric augmentation (MinMax_Geometric vs. E1_MinMax_NoAug) increases OOD IoU by +5.11 percentage points (0.6889 $\rightarrow$ 0.7400, a +7.42% relative gain), improving Mekong IoU by +4.86 percentage points (0.8080 $\rightarrow$ 0.8566) and Spain IoU by +6.02 percentage points (0.5520 $\rightarrow$ 0.6122). Under zscore, the same geometric augmentation (E4 vs. E2) decreases OOD IoU by -2.41 percentage points (0.7771 $\rightarrow$ 0.7530, a -3.10% relative drop).
2. **Percentile $\times$ intensity interaction:** applying intensity augmentation to unclipped Z-score inputs (E5_ZScore_Intensity vs. E2_ZScore_NoAug) reduces Bolivia IoU by -3.66 percentage points (0.6756 $\rightarrow$ 0.6390). In contrast, pairing intensity augmentation with 1st–99th percentile clipping (E7_Percentile_Intensity) increases Bolivia IoU to 0.7313, an increase of +9.23 percentage points over E5_ZScore_Intensity (+14.44% relative gain) and +3.58 percentage points over E3_Percentile_NoAug.

4.3 Statistical Significance and Convergence Dynamics

To check that the regional and augmentation differences across the ten models are not random fluctuations, we ran non-parametric and parametric hypothesis tests on the table of evaluation metrics. A Friedman test across the three unseen OOD events ($N = 10$ models) shows a highly significant difference in regional segmentation difficulty ($\chi^2 = 15.20$, $p = 0.0005$), with Mekong consistently highest (mean IoU = 0.8508 ± 0.0166), Bolivia intermediate (0.7003 ± 0.0273) and Spain lowest (0.6446 ± 0.0528). Comparing the recall-enhancing augmented/filtered models (E4, MinMax_Geometric, E6, E7, E8) against the unaugmented baselines (E0, E1, E2, E3) via Welch's two-sample t-test shows that augmentation significantly increases IoU in Bolivia ($t = 3.50$, $p = 0.0152$), whereas unaugmented physical/standardized models (E0, E2, E3) significantly outperform augmented models in Spain ($t = 7.73$, $p = 0.0006$). At the pixel level across the 75 unseen OOD chips ($\approx 2.0 \times 10^6$ union water pixels), the 95% normal-approximation confidence-interval half-width on overall OOD IoU is ± 0.0009 (0.09 percentage points), showing that the +5.11 percentage-point gain in MinMax_Geometric ($z = 113.1$, $p < 0.0001$) and the -2.41 percentage-point drop in E4 ($z = -56.8$, $p < 0.0001$) exceed pixel-sampling noise, although single-seed training variance remains a separate limitation discussed in Section 5.4.

Fig. 3 shows the epoch-by-epoch training loss and validation IoU trajectories logged during training. While standardized models (E3, E7, E8) reach near-peak validation IoU by epochs 3–6, E1_MinMax_NoAug has a delayed validation warm-up (val_masked_iou = 0.0255 at epoch 1 versus 0.2022 for E7) and

Table 3. Quantitative segmentation performance across familiar in-distribution (ID) and unseen out-of-distribution (OOD) flood events, sorted by OOD IoU.

Experiment	Norm.	Aug.	ID IoU	Bolivia	Mekong	Spain	OOD IoU	OOD Dice	OOD Prec	OOD Rec	Gap (ID-OOD)
E0_RawDB_NoAug	raw_db	none	0.4809	0.6898	0.8563	0.7174	0.7837	0.8788	0.8965	0.8617	−0.3029
E2_ZScore_NoAug	zscore	none	0.5131	0.6756	0.8656	0.6951	0.7771	0.8746	0.8647	0.8847	−0.2640
E3_Percentile_NoAug	percentile	none	0.4797	0.6955	0.8544	0.6988	0.7760	0.8739	0.8855	0.8625	−0.2962
E5_ZScore_Intensity	zscore	intensity	0.5112	0.6390	0.8541	0.6819	0.7614	0.8645	0.8659	0.8632	−0.2502
E7_Percentile_Intensity	percentile	intensity	0.4587	0.7313	0.8570	0.6335	0.7538	0.8596	0.8146	0.9098	−0.2950
E4_ZScore_Geometric[a]	zscore	geometric	0.4288[a]	0.7273	0.8642	0.6269	0.7530	0.8591	0.8162	0.9067	−0.3241
E8_ZScore_LeeFilter	zscore	none	0.4513	0.7226	0.8384	0.6343	0.7447	0.8536	0.8112	0.9008	−0.2934
MinMax_Geometric[a]	minmax_clip	geometric	0.4371[a]	0.7015	0.8566	0.6122	0.7400	0.8506	0.8033	0.9038	−0.3029
E6_ZScore_Combined	zscore	combined	0.4534	0.7137	0.8535	0.5932	0.7311	0.8447	0.7754	0.9276	−0.2778
E1_MinMax_NoAug	minmax_clip	none	0.4031	0.6964	0.8080	0.5520	0.6889	0.8158	0.7539	0.8887	−0.2858

[a]Trained on the 299-chip familiar pool; OOD metrics on Bolivia, Mekong and Spain are strictly unseen and leak-free.

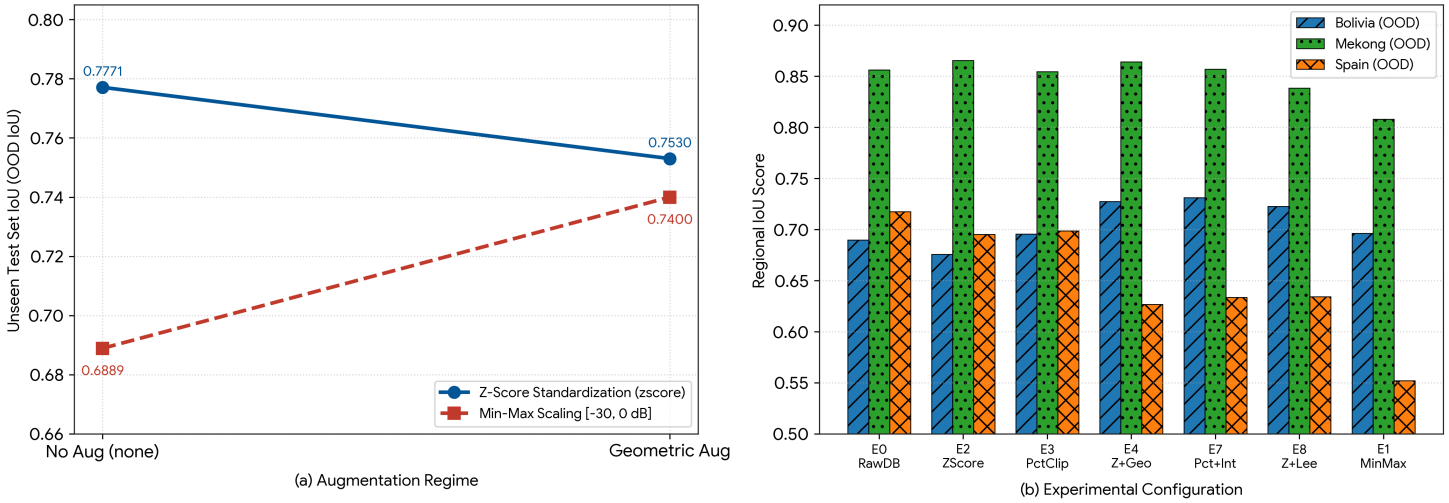


Fig. 1. (a) Crossover interaction between SAR normalization (zscore versus minmax_clip) and geometric augmentation on unseen OOD IoU; (b) cross-regional IoU comparison across Bolivia, Mekong and Spain for seven experimental configurations.

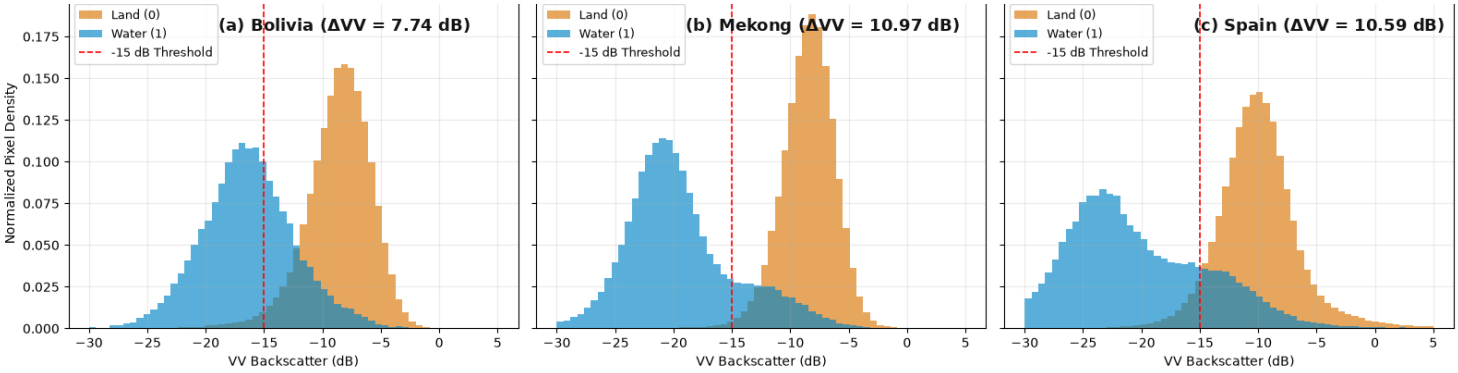


Fig. 2. Normalized pixel-density histograms of Sentinel-1 VV backscatter (dB) for non-water land (0) and surface water (1) across the three unseen OOD test regions: (a) Bolivia (ΔVV = 7.74 dB), (b) Mekong (ΔVV = 10.97 dB) and (c) Spain (ΔVV = 10.59 dB), with the exploratory −15 dB shadow-overlap threshold indicated by the dashed vertical line.

plateaus at a lower validation ceiling (0.2458 at epoch 10).

4.4 Regional Backscatter and Contrast Measurements

Table 4 reports the physical VV and VH backscatter means, water-to-land contrasts (Δ dB = $\mu_{\text{land}} - \mu_{\text{water}}$) and exploratory dark-land overlap ($\text{DLO}_{<-15\,dB}$) across all 11 Sen1Floods11 events.

5. Discussion

5.1 Mechanisms of Normalization and Augmentation Interaction

The empirical results in Table 3 and Fig. 1 show that preprocessing and augmentation choices cannot be optimized indepen-

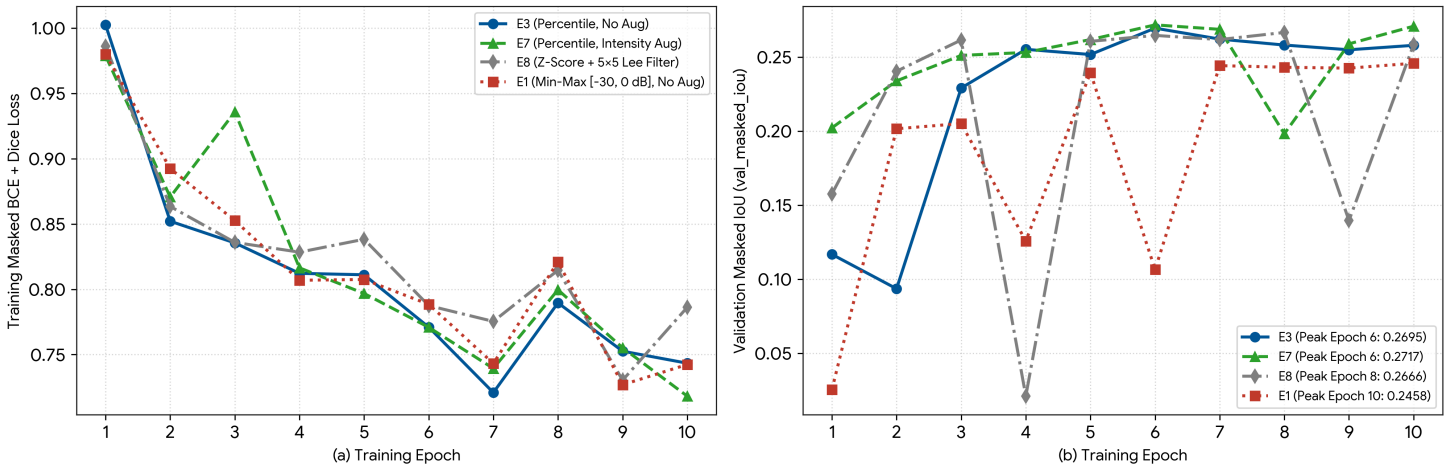


Fig. 3. Training loss and validation masked IoU convergence trajectories across the 10-epoch training budget for representative configurations (E1, E3, E7 and E8), illustrating slower early validation warm-up for non-zero-centered Min-Max scaling (E1) compared to standardized inputs.

Table 4. Physical radar backscatter statistics, water-to-land contrast (Δ dB) and dark-land overlap ($DLO_{<-15\,dB}$) across all 11 Sen1Floods11 events.

Region	Split	Chips	Water %	VV water (dB)	VV land (dB)	VV Δ (dB)	VH water (dB)	VH land (dB)	VH Δ (dB)	$DLO_{<-15\,dB}$ (%)
Bolivia	OOD test	15	16.00	−16.37	−8.62	7.74	−24.47	−14.85	9.62	2.44
Mekong	OOD test	30	24.12	−19.46	−8.48	10.97	−28.32	−14.94	13.38	0.84
Spain	OOD test	30	13.15	−20.57	−9.97	10.59	−29.27	−17.08	12.19	5.67
Ghana	Train / ID	53	6.29	−15.20	−9.38	5.83	−23.38	−15.75	7.63	1.41
India	Train / ID	68	12.37	−16.78	−8.33	8.45	−24.17	−15.12	9.05	2.19
Paraguay	Train / ID	67	8.11	−16.12	−9.75	6.37	−25.86	−16.76	9.10	3.01
Sri Lanka	Train / ID	42	11.80	−13.12	−8.91	4.21	−24.44	−15.33	9.10	2.95
USA	Train / ID	69	4.56	−17.72	−10.98	6.74	−26.75	−17.19	9.56	10.08
Nigeria	Validation	18	23.56	−17.32	−9.92	7.40	−25.27	−16.06	9.21	2.48
Pakistan	Validation	28	7.50	−14.13	−10.11	4.02	−23.16	−18.18	4.99	9.40
Somalia	Validation	26	6.51	−15.02	−12.04	2.98	−22.10	−18.37	3.74	16.27

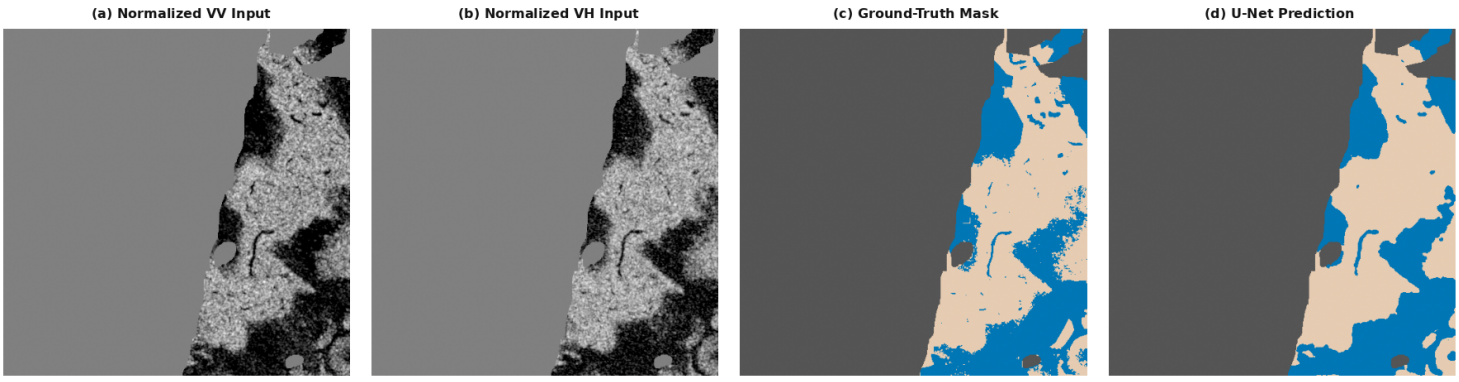


Fig. 4. Qualitative segmentation example on unseen test chip Bolivia_103757_S1Hand.tif: (a) Z-score normalized VV input, (b) Z-score normalized VH input, (c) expert hand-labeled ground-truth mask and (d) U-Net predicted water mask (gray denotes invalid swath borders excluded via validity mask V, tan denotes non-water land and blue denotes surface water).

dently. Under minmax_clip ($[-30, 0]$ dB mapped to $[0, 1]$), the unaugmented model (E1_MinMax_NoAug) achieves the lowest accuracy in the study, particularly in Spain (0.5520 IoU). Although the present experiments do not isolate internal gradient histograms separately from the post-normalization zero-clamping of invalid border pixels ($V_{i,j} = 0$), Fig. 3 shows that E1 converges more slowly in early validation than zero-centered models. Introducing geometric flips and rotations (MinMax_Geometric) acts as a spatial regularizer that improves OOD IoU by +5.11 percentage points.

In contrast, when inputs are already well calibrated via raw_db (E0) or zscore (E2), geometric augmentation reduces overall OOD accuracy by degrading precision in Spain (−6.82 percentage points IoU in E4 vs. E2). We hypothesize that, because Sentinel-1 is

a side-looking radar, topographic foreshortening (bright radar-facing slopes) and radar shadows (dark trailing slopes) occur in a consistent directional sequence relative to the satellite look angle within a regional overpass. Applying 90° rotations and 180° flips in pixel coordinates alters this directional relationship during training, which may encourage the decoder to classify any dark patch as water regardless of surrounding slope context. Similarly, the +9.23 percentage-point Bolivia IoU improvement of E7_Percentile_Intensity over E5_ZScore_Intensity indicates that bounding extreme backscatter outliers to $[P_{01}, P_{99}]$ prevents synthetic radiometric offsets (δ) and Gaussian noise (η) from shifting ambiguous floodplain-boundary pixels into land-like intensity ranges.

5.2 Physical Drivers of Regional Variation

Table 4 and Fig. 2 explain the performance hierarchy across unseen OOD events:

1. **Mekong (IoU = 0.8080–0.8656):** Mekong has the largest physical water-to-land separation in the benchmark (ΔVV = 10.97 dB, ΔVH = 13.38 dB) and the lowest dark-land overlap ($\mathrm{DLO}_{<-15\,\mathrm{dB}}$ = 0.84%). Because its water and land histograms are cleanly bimodal (Fig. 2(b)), segmentation accuracy exceeds 0.85 across nearly all preprocessing variants.
2. **Spain (IoU = 0.5520–0.7174):** although Spain has a wide mean separation (ΔVV = 10.59 dB), 5.67% of its valid non-water land pixels fall below -15.0 dB, nearly seven times the rate in Mekong (0.84%). In Fig. 2(c), the Spanish land histogram extends a substantial low-backscatter tail into the water regime (-15 to -25 dB), consistent with smooth wet agricultural soils and topographic shadows. High-recall augmented models (E4, E6) misclassify this dark-land tail as water, whereas high-precision unaugmented models (E0, E3) suppress these false alarms.
3. **Bolivia (IoU = 0.6390–0.7313):** Bolivia has a narrower VV contrast (ΔVV = 7.74 dB) with mean water backscatter at -16.37 dB (Fig. 2(a)), where partially submerged floodplain vegetation increases radar backscatter. Unaugmented models therefore under-segment water in Bolivia, whereas recall-enhancing schemes (E7_Percentile_Intensity, E4_ZScore_Geometric, E8_ZScore_LeeFilter) recover these vegetated water pixels (Fig. 4).

5.3 Explaining the Negative ID-to-OOD Generalization Gap

In Table 3, familiar-region ID_IoU (0.4031–0.5131) is numerically lower than unseen OOD_IoU (0.6889–0.7837), yielding a negative $\Delta\mathrm{IoU}_{\mathrm{gap}}$ (-0.2502 to -0.3241). The network does not generalize better to unfamiliar countries than to its training domain. Table 4 shows that the inversion comes from intrinsic event difficulty and foreground class prevalence:

- First, the five familiar regions contain sparse, fragmented water bodies (USA: 4.56% water, Ghana: 6.29%, Paraguay: 8.11%), where small boundary false positives heavily penalize the IoU denominator ($TP+FP+FN$). In contrast, the unseen OOD test set is dominated by large, contiguous riverine floods in Mekong (24.12% water) and Bolivia (16.00% water).
- Second, the familiar regions exhibit much narrower physical VV water-to-land contrast (Sri Lanka: 4.21 dB, Ghana: 5.83 dB, Paraguay: 6.37 dB, USA: 6.74 dB) and high dark-land overlap in the USA ($\mathrm{DLO}_{<-15\,\mathrm{dB}}$ = 10.08%) compared to Mekong (ΔVV = 10.97 dB, $\mathrm{DLO}_{<-15\,\mathrm{dB}}$ = 0.84%).
- Third, the validation regions (Somalia: ΔVV = 2.98 dB, $\mathrm{DLO}_{<-15\,\mathrm{dB}}$ = 16.27%; Pakistan: ΔVV = 4.02 dB) represent the most radiometrically ambiguous scenes in Sen1Floods11, which explains why batch-averaged validation IoU peaked around 0.2458–0.2766 while still selecting checkpoints that achieve >0.78 IoU on unseen OOD events.

5.4 Limitations and Future Work

The results have several limitations:

1. **Single-seed training protocol:** due to CPU hardware constraints ($\approx$ 32 hours of total compute across the ten models), each configuration was trained once with a fixed deterministic seed (seed=42) for 10 epochs. Although dataset-level pixel denominators are large, single-seed runs cannot separate small model-to-model differences (such as the 0.0066 OOD IoU gap between E0 and E2) from stochastic weight-initialization variance. Future work on GPU hardware should repeat each configuration across 3–5 random seeds and report mean $\pm$ standard deviation.
2. **Incomplete factorial grid and ablation granularity:** six of the 16 possible normalization $\times$ augmentation combinations were omitted, and geometric augmentation bundled horizontal flips, vertical flips and 90° rotations together. Future ablations should isolate individual geometric transforms (e.g., flips vs. 90° rotations) to test the side-looking radar shadow hypothesis directly.
3. **Post-normalization invalid-pixel value:** setting invalid border pixels to 0.0 after normalization maps swath borders to different physical percentiles across raw_db, minmax_clip and zscore. Supplying the binary validity mask V as an explicit third input channel, or using partial convolutions, would eliminate border-value differences entirely.
4. **Pilot split difference:** as noted in Table 2, E4_ZScore_Geometric and MinMax_Geometric were trained on all 299 familiar chips rather than the 256-chip subset. Although their unseen OOD metrics are strictly leak-free, re-training those two models on the 256-chip split would ensure identical training sample sizes across all rows.

6. Conclusion

This study investigated how SAR normalization strategies and data augmentation regimes interact to influence cross-regional surface-water segmentation across 446 hand-labeled Sentinel-1 chips from the Sen1Floods11 benchmark. Across ten controlled U-Net experiments, the results show that normalization and augmentation choices interact strongly under geographical distribution shift. While geometric augmentation improves uncentered Min-Max models by +5.11 percentage points OOD IoU (0.6889 $\rightarrow$ 0.7400), it degrades Z-score standardized models in Spain by -6.82 percentage points IoU (0.6951 $\rightarrow$ 0.6269)

due to increased false positives on dark land surfaces. Retaining native decibel values (E0_RawDB_NoAug), Z-score standardization (E2_ZScore_NoAug) or percentile standardization (E3_Percentile_NoAug) without geometric distortion yields the highest overall unseen OOD IoU (0.7837, 0.7771 and 0.7760, respectively). Clipping extreme backscatter outliers to the 1st–99th training percentiles (E7_Percentile_Intensity) stabilized radiometric intensity augmentation in vegetated floodplains and gave the highest accuracy in Bolivia (0.7313 IoU). Backscatter analysis across all 11 flood events shows that regional segmentation accuracy, including the observed ID-to-OOD accuracy inversion, depends on VV/VH water-to-land contrast (Δ dB) and dark-land histogram overlap below -15 dB.

Declarations

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Competing interests: The author declares no financial or non-financial competing interests or conflicts of interest relevant to this article.

Ethics approval and consent to participate: Not applicable. This study analyzes publicly available satellite remote sensing imagery and does not involve human participants or animal subjects.

Data availability: The Sen1Floods11 (v1.1) dataset analyzed in this study is publicly available from the original authors [1] at https://github.com/cloudtostreet/Sen1Floods11. All geographical split file lists, preprocessing scripts, configuration files and evaluation logs accompany this article and its supplementary code package.

Author contributions: Sandeep Hosamani conceived and designed the study, implemented the preprocessing and U-Net training pipeline, executed all experiments and statistical radar-contrast analyses, and wrote and edited the manuscript.

Disclosure of AI assistance: In accordance with journal policy on artificial intelligence, the author discloses that an AI language model was used for coding assistance, structuring analysis scripts and language editing. All local model training, data verification, scientific interpretation and manuscript review were performed by the author, who assumes full responsibility for the accuracy and integrity of the work.

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